

Weekly Engineering & Product Report

Comprehensive 7-Day Implementation Summary & Architectural Progress

REPORTING CYCLE

Sep 08 – Sep 15, 2026 (7 Days)

PLATFORM ARCHITECTURE

Go Digital CRM • Multi-Tenant

VERIFICATION HEALTH

170+ Test Suites Passing

PRODUCTION COMMITS

10 Major

Web, Edge, DB & Mobile

DATABASE MIGRATIONS

12 Applied

100% applied & tested

FILES MODIFIED

120+ Files

Over 8,500 LOC updated

VERIFICATION RATE

100% Pass

Zero schema/lint errors

1. Executive Overview & Strategic Milestones

High-level synthesis of features, workflow stabilization, and operational impact delivered this week.

During the 7-day sprint from September 08 through September 15, 2026, engineering delivered critical multi-channel communication tools (TeleCMI Cloud Telephony agent provisioning and personal WhatsApp socket gateway), reinforced lead lifecycle pipelines, and deployed AI-powered sales enablement capabilities.

Crucial operational friction points were permanently unblocked: the Telecaller task workspace query was restored from runtime Zod crashes, lead status queue double-counting between Contacted and Follow-up was resolved, and mobile Expo dependencies were updated to SDK 57 patches, achieving a clean 21/21 doctor score.

2. Key Modules & Functional Capabilities Implemented (Part 1)

Production capabilities delivered for Cloud Telephony and WhatsApp messaging infrastructure.

A. TeleCMI Cloud Telephony & Audited Sales Handoff Workflows

Voice & Telephony

- TeleCMI Agent Provisioning UI & Edge Function: Built admin interface (telecmi-agent-editor.tsx) and Edge Function (integration-telecmi-provision-agent) allowing dealership administrators to create, configure, and assign TeleCMI agent credentials and extension lines directly from the CRM.
- Audited Sales Handoff Cancellation: Implemented SQL procedure (202609120001_cancel_telecaller_sales_handoff.sql) allowing telecallers/team managers to safely cancel accidental or invalid lead handoffs, returning leads to telecalling with full audit history.
- Mandatory Handoff Reasons: Enforced required reason recording (202609100002_require_sales_handoff_reason.sql) so no lead can be transitioned to sales without clear justification.
- Calls Query Performance Optimization: Deduplicated and hoisted scope checks in legacy calls workspace (202609090003 & 202609090004), evaluating access once per (branch, team, owner) triple and slashing scope checks from 123 to 3 in benchmarks.

B. Personal WhatsApp Gateway & Live Tunnel Infrastructure

Real-Time Messaging

- Baileys 7.0.0-rc14 Socket Gateway: Integrated standalone Node service managing multi-device WhatsApp connections with generation fencing, 45s socket leases, and AES-256-GCM encrypted Signal keys.
- Automated HTTPS Cloudflare Tunnel: Created local-https.mjs development orchestration that boots the gateway, launches a secure Cloudflare tunnel, and automatically synchronizes the tunnel URL & signing secrets into Supabase Edge Functions.
- Text History Sync Engine: Implemented personal-whatsapp-sync Edge Function & database schema to fetch up to 30 days of text history safely without contact harvesting or unlinked conversation leaks.
- Attention Queue Direct Actions: Added verified customer phone numbers directly to priority dashboard attention cards (202609080009_dashboard_attention_phone.sql), enabling 1-click calling and WhatsApp chat initiation.

2. Key Modules & Functional Capabilities Implemented (Part 2)

Operations hardening, AI intelligence, Customer 360, and mobile ecosystem improvements.

C. Lead Lifecycle & Telecaller Queue Hardening

CRM Operations

- **Mandatory Form Validation UX:** Added explicit red asterisks to the 4 essential lead creation fields (Customer Name, Phone Number, Lead Source, Branch) to eliminate confusing client-side validation failures before submission.
- **Lead-Anchored Task Raising:** Integrated task creation directly with lead rows (`?action=create&lead=...`), eliminating orphan unassigned tasks and preserving full customer context.
- **Contacted vs Follow-up Pipeline Ladder:** Fixed status collision (`202609090001_contacted_queue_excludes_open_followups.sql`) so leads with scheduled follow-ups do not double-count under both Contacted and Follow-up tabs.
- **Telecaller Tasks Workspace Recovery:** Restored three-stage query architecture (scoped KPIs -> searched tab counts -> authorized rows), eliminating Zod parsing crashes and restoring accurate overdue counters.
- **Inline Follow-Up Resolution:** Prevented dead-end UI toasts by opening an inline resolution dialog for blocked actions, with instant "Complete", "Cancel", and automatic action replay.
- **Accurate CSV Import Diagnostics:** Replaced generic "validation rejected" errors with precise diagnosis (detecting un-deployed Edge Functions vs malformed rows).

D. AI-Powered Vehicle Comparison & Sales Intelligence

AI & Sales Tech

- **AI Vehicle Comparison Engine:** Launched vehicle-comparison-ai Edge Function delivering automated side-by-side spec comparisons, vehicle advantages, and talking points for sales consultants.
- **Competitor Comparison Workspace:** Interactive matrix in sales module displaying multi-brand comparisons (`competitor-comparison-workspace.tsx`) with instant AI summary card generation.
- **Public Shareable Comparison Links:** Secure tokenized public access (`202609080005_shared_vehicle_comparison.sql`) enabling sales reps to share branded vehicle spec sheets with customers via WhatsApp.
- **Pricing Defaults & Targets:** Implemented variant pricing defaults and branch model sales target configuration tables.

E. Customer 360, In-App Reminders & Mobile Upgrades

UX & Platform

- **Customer Additional Metadata Fields:** Built flexible schema and UI for dealership-defined customer attributes (`202609080002_customer_additional_fields.sql`).
- **Follow-Up Web Reminders:** Built real-time in-app notification manager (`followup-reminder-manager.tsx`) and Zustand store alerting consultants when follow-up callbacks or appointments are due.
- **Global Credit Balance Header:** Added real-time SMS, WhatsApp, and AI credit balance display directly into top app header.
- **Mobile SDK 57 Patch Updates:** Bumped expo (57.0.21) and expo-router (57.0.20), passing all 21/21 expo-doctor diagnostic checks and maintaining strict mobile type safety.

Architectural Safeguards & Tenancy Enforcement

- **Strict Tenancy Isolation:** All tables enforce `organization_id` and `branch_id` boundaries under Postgres Row-Level Security (RLS). No client query can bypass assigned role ceilings.
- **Audited Mutation Logs:** Handoff cancellations, lead status transitions, and agent provisioning log immutable audit entries to maintain complete dealership accountability.
- **Zero Plaintext Exposure:** Baileys Signal keys and TeleCMI tokens use server-side AES-256-GCM encryption with generation fencing, ensuring zero secrets leak to the frontend.

3. Database Migrations Applied (12 Applied in 7 Days)

Complete record of database schema updates, RLS policies, and performance indexes deployed.

MIGRATION FILENAME	PURPOSE & ARCHITECTURAL IMPACT	DOMAIN
202609120001_cancel_telecaller_sales_handoff.sql	Audited sales handoff cancellation workflow and state recovery	Telephony & Leads
202609100002_require_sales_handoff_reason.sql	Mandatory reason enforcement for telecaller lead transfers	Audit & Compliance
202609100001_optimize_marketing_performance_scope.sql	Query performance & RLS optimization for marketing metrics	Analytics & RLS
202609090004_calls_workspace_scope_hoist.sql	Hoisting scope evaluation across distinct branch/team triples	Performance
202609090003_calls_workspace_scope_dedupe.sql	Deduplicated repeated lateral permission checks in calls workspace	Performance
202609090002_telecaller_task_page_status_counts.sql	Restored 3-stage query architecture for telecaller task KPIs & status counts	Tasks & Work
202609090001_contacted_queue_excludes_open_followups.sql	Mutually exclusive lead queue logic between Contacted & Follow-up	Lead Queues
202609080009_dashboard_attention_phone.sql	Direct phone number visibility on dashboard attention cards	Dashboard UX
202609080008_followup_web_reminders.sql	Active lead reminder triggers & browser notification states	Notifications
202609080005_shared_vehicle_comparison.sql	Tokenized public access links for client vehicle spec comparisons	Sales AI
202609080004_variant_pricing_defaults.sql	Default pricing models and branch model targets schema	Inventory
202609080002_customer_additional_fields.sql	Configurable custom schema fields for dealership customer 360	Customer 360

4. Weekly Git Commit Chronology (Sep 08 – 15, 2026)

Audit log of primary commits pushed to the repository during this weekly cycle.

HASH	DATE	SCOPE	COMMIT DESCRIPTION & SYSTEM IMPACT
545daec	Sep 12	Chore	Update Next.js type reference paths to dev directory
781ebb0	Sep 12	Feature	Add TeleCMI agent provisioning and sales handoff cancellation workflows
6d0b86a	Sep 10	Fix	Report real reason when lead CSV import fails (differentiate 404 vs bad row)
f6a5866	Sep 09	Feature	Mark required lead fields with red asterisks and raise tasks from lead rows
a7a000b	Sep 09	Perf	Cut redundant scope checks and hoist evaluations in legacy calls query
1b1b7b4	Sep 09	Chore	Bump Expo (57.0.21) and Expo Router (57.0.20) SDK 57 patch versions
8c72f0c	Sep 09	Fix	Align telecaller queues to status ladder and unbreak tasks page schema
c3b17fc	Sep 09	Chore	Add WhatsApp gateway HTTPS tunnel dev script with Edge secrets sync
6d4c954	Sep 08	Feature	Add phone numbers to attention queue items for 1-click calling & WhatsApp
e8beaab	Sep 08	Feature	Integrate WhatsApp sync, AI vehicle comparison, reminders & Customer 360

5. Upcoming Focus Areas for Next Sprint

Strategic priorities scheduled for the following 7-day engineering iteration.

- WhatsApp Pilot Group Scanning: Roll out linked device QR scanning to pilot telecaller cohort; monitor memory/RSS and socket lease recovery.
- TeleCMI End-to-End Live Calls: Validate live browser dialing, incoming caller screen popups, and automatic call recording storage.
- Full Customer Journey Walkthrough: Execute end-to-end integration test from lead qualification to delivery and feedback collection.
- Outbound Messaging Temperature Queue: Finalize HOT -> WARM -> COLD strict priority ordering on bulk marketing campaigns.